INCH-POUND
MIL-I-24768/31
25 February 1992

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, GLASS-MAT, POLYESTER-RESIN (GPO N-1P)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a mat of random-laid glass fibers which is saturated with a polyester-resin combined with suitable fillers, cured by heat and pressure and processed to conform to this specification.

Intended use: The laminated material covered by this specification is a rigid plastic material intended for general-purpose electrical equipment applications requiring good dielectric properties and good punching characteristics.

Forms: Sheets are the only recommended form of laminated material covered by this specification.

Temperature classification: The laminated material covered by this specification has a temperature classification of class 130.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/31-S-1	0.062	0.008
M24768/31-S-2	0.094	0.009
M24768/31-S-3	0.125	0.010
M24768/31-S-4	0.156	0.011
M24768/31-S-5	0.188	0.012
M24768/31-S-6	_[1]	_[2]

[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/31-S-6-0.14.

[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Performance requirements: Performance requirements for laminated sheets shall be as specified in table II.

The following performance requirements do not apply to this plastic laminate:

a. Volume resistivity
b. Surface resistance
c. Dielectric strength
d. Specific gravity
e. Compressive strength
f. Flammability.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

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TABLE II. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]					
				0.062	0.125	0.188	0.500	1.000	1.001 – 2.000
Dielectric breakdown voltage	A D-48/50	4.6.4	Min. kV	40.0 5.0	40.0 5.0	40.0 5.0			
Dielectric constant									
60 Hz:	A D-24/23	4.6.6	Max.		6.0 6.5				
1 MHz:	A D-24/23				5.0 6.0				
Dissipation factor									
60 Hz:	A D-24/23	4.6.6	Max.		0.10 0.15				
1MHz:	A D-24/23				0.03 0.06				
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm						
Surface resistance	C-96/35/90	4.6.8	Min. megohm						
Arc resistance	A D-48/50	4.6.9	Min. sec.	100 100	100 100	100 100			
Tracking resistance	A	4.6.10	Min. minutes						
Impact strength	E-48/50	4.6.11	Min. ft-lbs. per inch		5.0	5.0			
Flexural strength	A	4.6.12	Min. psi	16,000	16,000	16,000			

See footnotes at end of table.

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TABLE II. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]					
				0.062	0.125	0.188	0.500	1.000	1.001–2.000
Thermal endurance Flexural strength:	E-1/130: T-130	4.6.13	Min psi	9,000	9,000	9,000	9,000	9,000	9,000
Bonding strength	A D-48/50	4.6.15	Min. pounds						
Water absorption	D_1-24/23	4.6.17	Max. percent	.1.0	.0.7				
Silicone content	E-168/185	4.6.18	Max. ppm						
Flame resistance Ignition time:	A	4.6.19	Min. sec.						
Burning time:			Max. sec.						

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.

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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
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